

ECE 251C – Fall 2025 – Parhi
Quiz 2

Quiz Rules: No notes, computers, calculators, or phones; have only a pen/pencil; work independently.

Please show your work and provide any justification of your answers so that you can receive partial credit.

On my honor, I pledge that I have neither given nor received unpermitted aid on this exam.

signature:

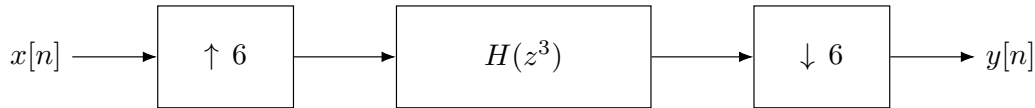
name:

(a) _____ / 15 points

(b) _____ / 10 points

Total: _____ / 25 points

Consider the multirate system



(a) Find $Y(z)$ in terms of $X(z)$ and $H(z)$.

Solution. *We have that*

$$\begin{aligned}
 & \rightarrow \boxed{\uparrow 2} \rightarrow \boxed{\uparrow 3} \rightarrow \boxed{H(z^3)} \rightarrow \boxed{\downarrow 3} \rightarrow \boxed{\downarrow 2} \rightarrow \\
 \equiv & \rightarrow \boxed{\uparrow 2} \rightarrow \boxed{\uparrow 3} \rightarrow \boxed{\downarrow 3} \rightarrow \boxed{H(z)} \rightarrow \boxed{\downarrow 2} \rightarrow \\
 \equiv & \rightarrow \boxed{\uparrow 2} \rightarrow \boxed{H(z)} \rightarrow \boxed{\downarrow 2} \rightarrow
 \end{aligned}$$

At this point, there are multiple ways to proceed. One option is to simplify the above system in the Fourier domain to find

$$Y(e^{j\omega}) = \frac{1}{2} \left(H(e^{j\omega/2}) + H(e^{j\omega/2+\pi}) \right) X(e^{j\omega}). \quad (1)$$

In which case, we have that

$$\boxed{Y(z) = \frac{1}{2} \left(H(z^{1/2}) + H(-z^{1/2}) \right) X(z).} \quad (2)$$

Alternatively, one could also notice that the last system in the simplification above is equivalent to filtering $x[n]$ with only the even coefficients of $h[n]$. If the polyphase representation of $H(z)$ is $H(z) = H_{\text{even}}(z^2) + z^{-1}H_{\text{odd}}(z^2)$, this implies that

$$\boxed{Y(z) = H_{\text{even}}(z)X(z).} \quad (3)$$

Both of these are correct and valid answers.

The following are possible point breakdowns for this question:

- 15/15: Completely correct derivation via either approach (or a different one) described above.
- 8/15: Correct simplification of the multirate system and something simple went wrong.
- 5/15: Correct or almost correct simplification of the multirate system, but ended up with something completely off for $Y(z)$ in terms of $H(z)$ and $X(z)$.
- 0/15: Problem left blank or approach is completely wrong.

- (b) Sketch $|Y(e^{j\omega})|$ when $H(z) = -1 + 9z^{-2} + 16z^{-3} + 9z^{-4} - z^{-6}$ and $x[n] = (-1)^n$.

Solution. From either of the boxed solutions of (a), we see that the system is equivalent to filter $x[n]$ with the even coefficients of $h[n]$. Observe that the $h[n]$ is Type-I linear-phase and so $h_{\text{even}}[n] = h[2n]$ is Type-II linear-phase. Type-II linear-phase filters always have a zero at π and $x[n] = e^{j\pi n}$ is a pure frequency at π (recall that its DTFT is a Dirac comb). Therefore $(h_{\text{even}} * x)[n] = 0$ for all n . In other words, $|Y(e^{j\omega})|$ is identically zero so its plot is simply that.

The following are possible point breakdowns for this question:

- 10/10: Completely correct identification of the zero spectrum either from the argument above or a different one (e.g., a brute force computation or via sketches).
- 7/10: A valid approach to determining $|Y(e^{j\omega})|$ via computations or sketches, but a simple error occurred along the way.
- 5/10: A valid approach to determining $|Y(e^{j\omega})|$ via computations or sketches, but a major error occurred along the way.
- 0/10: Problem left blank or approach is completely wrong.